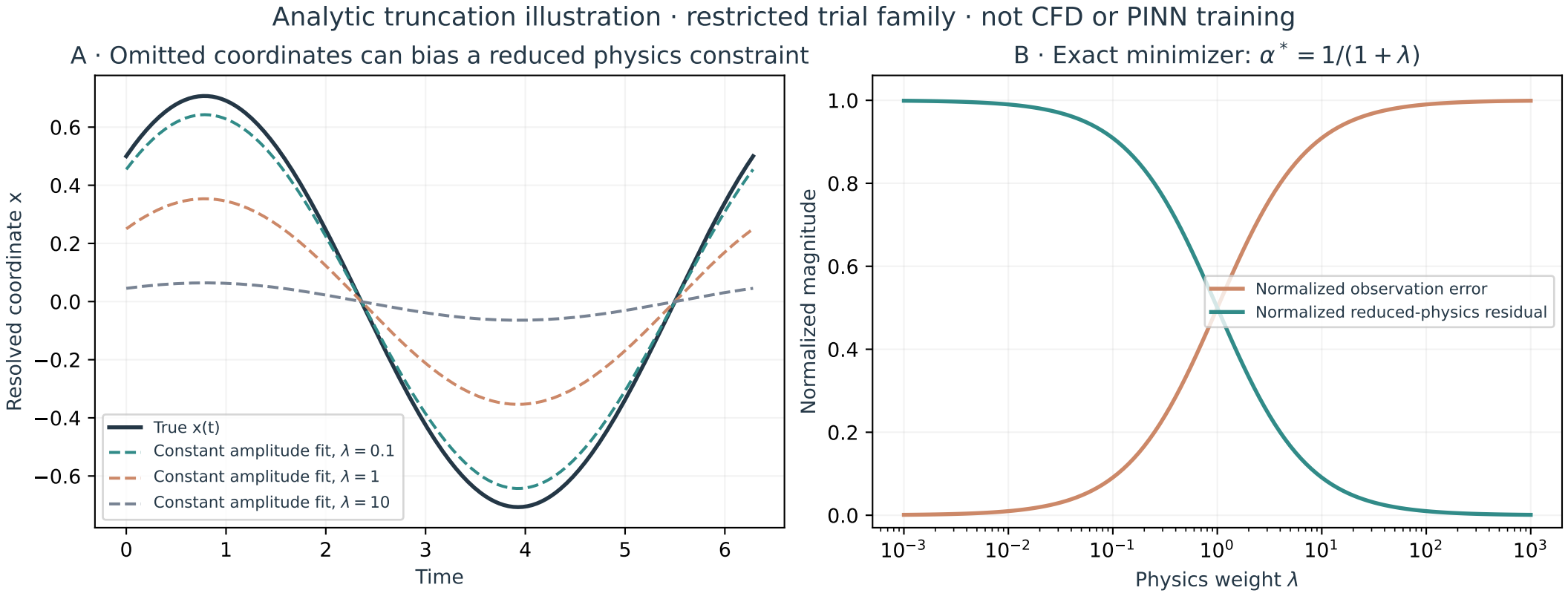
Analytic truncation illustration · restricted trial family · not CFD or PINN training
A · Omitted coordinates can bias a reduced physics constraint
B · Exact minimizer: $\alpha^* = 1/(1+\lambda)$
Resolved coordinate x
Time
True x(t)
Constant amplitude fit, $\lambda = 0.1$
Constant amplitude fit, $\lambda = 1$
Constant amplitude fit, $\lambda = 10$
Normalized magnitude
Physics weight λ
Normalized observation error
Normalized reduced-physics residual